

Diffusion-Limited Crystal Growth of Gallium Nitride Using Active Machine Learning

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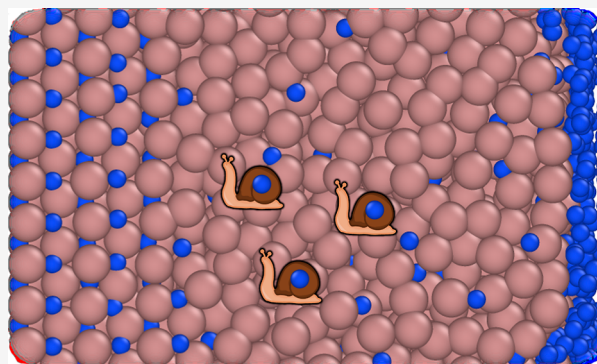


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ABSTRACT: Gallium nitride (GaN) is an important semiconductor with properties that make it particularly suitable for high-temperature applications in power systems. Unfortunately, its crystalline synthesis involves an energy-intensive process requiring high temperatures and pressures. A new additive manufacturing process offers a lower-temperature alternative, but little is understood of the molecular-scale mechanisms that drive its crystallization from the melt. Traditional semiempirical force fields within a molecular dynamics (MD) simulation are typically unable to capture bond-making and -breaking, whereas *ab initio* MD, while more accurate, suffers from high computational expense. This paper uses a machine-learned force field based on the FLARE++ framework to mimic the liquid-phase epitaxy of GaN, simulating the diffusion of nitrogen atoms through liquid gallium to form GaN. We show that nitrogen diffusion through Ga is slow, and relatedly, there is a pronounced tendency for nitrogen to phase-segregate within liquid Ga. This leads to nitrogen being less available to react and form crystalline GaN. As a result, the predicted crystal growth at the melt/crystal interface is extremely slow, as seen experimentally. In this work, we demonstrate the potential of a MLFF to describe complex, multiphase behavior under different conditions. We also uncover the key atomistic-level mechanism of diffusion-limited GaN crystal growth, which is an important step toward further control of the additive manufacturing process.



1. INTRODUCTION

Considerable research has been dedicated to developing the next generation of semiconductor materials to complement or, in certain applications, even replace silicon-based semiconductors. The search continues for materials with better electronic performance, more versatile thermomechanical properties, and cheaper synthesis for economical production.^{1–4} Research since the 1960s has investigated III–V materials, such as GaAs, which display better carrier mobility than silicon and a direct band gap.⁵ Gallium nitride (GaN) has emerged over the past few decades as an excellent choice of material for power and optoelectronic devices with its wide band gap, high mobility, and high saturation velocity.⁶ The wide band gap opens the door to high-temperature operation and other benefits. However, the performance of current devices is still far from the theoretical limit for GaN,⁷ with large potential for improvement if defects can be further controlled via novel fabrication methods.

GaN has been deployed in many industrial applications, such as aerospace, defense, and small electronic devices.^{8–10} More recently, GaN is being considered a prime contender for cutting the recharging time for electric vehicles.¹¹ Although GaN has superior electronic and thermal performance compared to traditional silicon devices, energy-intensive manufacturing and the relatively small scale of current GaN

manufacturing processes currently deter the widespread use and affordable application of GaN in consumer electronics.¹² Modern GaN manufacturing typically uses molecular beam epitaxy or chemical vapor deposition (CVD), both of which are energy-intensive.¹³ Homoepitaxy produces high-quality films with low defect densities, but such substrates are very expensive. As stated by Menenghini et al.,⁶ “bulk GaN crystal growth process is highly challenging, requiring high temperatures above 2200 °C and nitrogen pressures above 6 GPa, making it impossible to crystallize GaN from the melted compound.”¹⁴ As a result, there is considerable incentive to improve current synthesis methods to grow high-quality crystalline GaN under near-ambient conditions.

As reported by Gagnon et al., researchers at the Johns Hopkins Applied Physics Laboratory have developed an additive manufacturing method to produce GaN that allows epitaxial crystal growth in liquid gallium using gaseous

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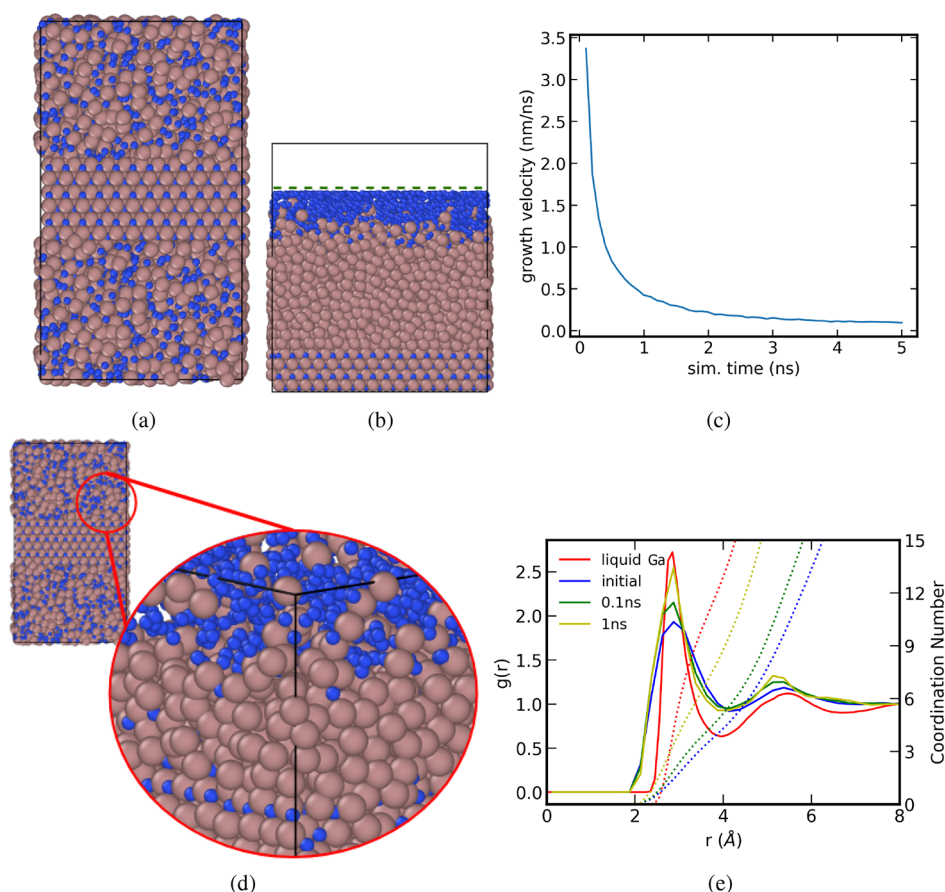


Figure 1. Simulation and analysis of results from simulations performed using the FLARE force field within LAMMPS at 2000 K. (a) Illustration of the “sandwich” setup. The substrate surface exposes the *c*-plane of the wurtzite structure. Gallium atoms are shown as light brown spheres and nitrogen atoms as blue spheres throughout. Black lines mark the simulation box’s rectangular periodic boundaries. (b) Simulation setup used to mimic nonperiodic growth, also at 2000 K. The exposed substrate surface is the wurtzite *c*-plane. The green dashed line shown represents the location of the repulsive wall imposed to prevent atoms from crossing the periodic boundary in the *z*-direction. (c) Growth velocity as a function of simulation time for the “sandwich” setup as shown in (a). Initial velocity was estimated to be 3.37 m/s; average velocity over the last 1 ns of the simulation was just 0.11 m/s. (d) Evidence of phase separation (clustering of N atoms) in the melt region. (e) RDF, $g(r)$, of Ga–Ga pairwise interactions in the melt region, illustrated after phase separation had started to occur. $g(r)$ were calculated after 0 (blue solid line), 0.1 (green solid line), and 1 ns (yellow solid line) of simulation time. $g(r)$ of liquid gallium is shown as a red solid line.²⁶ The cumulative coordination numbers were calculated with the label on the right-hand side of the *y*-axis. They have the same color code as the corresponding $g(r)$ as dotted lines.

ammonia as the nitrogen source.¹⁵ This new approach significantly reduces the energy cost because it operates under ambient pressure and temperatures around 1000–1500 K, conditions much milder than other common growth techniques for GaN. Additive manufacturing also enables fast prototyping and more customizable semiconductor design.^{16–18} However, the underlying reaction mechanisms remain unknown, including the nature of the diffusion process of the atomized nitrogen species and their subsequent reaction with liquid gallium droplets. Just as important, there is a need to establish a growth model to quantify the growth of GaN crystalline structures on both its polar and nonpolar surfaces. This review will help to uncover some of these mechanisms.

Previous modeling work on GaN crystallization has focused on understanding CVD and molecular beam epitaxy processes, not solidification from the melt.^{19,20} In addition, there has been no atomic-scale modeling of adatom diffusion on a substrate surface due to the limited force fields capable of modeling both the bulk/melt regions and adequately representing all the different phases and atom-types involved in the process. Even though there has been previous work studying the crystallization of liquid semiconductors and associated energy

changes, the system was constrained to bulk phase solidification from liquid to crystal.^{21–23} Therefore, there remains a need to understand the reaction mechanism of additive manufacturing processes for semiconductors with an accurate and transferable force field capable of representing different phases. To do so, we utilize a machine learning-derived transferable force field called FLARE,^{24,25} that is capable of capturing a reaction-diffusion system and modeling several chemical species and the multiple phases involved in this additive manufacturing process.²⁶ In this work, we demonstrate how we apply this actively learned force field to model the GaN crystal growth process. Mimicking the additive manufacturing process, we will provide an analysis of the nitrogen diffusion and its subsequent reaction with a gallium melt at ambient pressure and temperatures around 1000 °C to produce GaN. Lastly, we investigate the preference of certain planes (surface terminations) on which crystal formation is more likely to happen and explain how the choice of plane affects the resulting smoothness of the film.

2. COMPUTATIONAL METHODS

In a previous paper, we discussed the details of a force field that we developed that was capable of combining models for the elemental constituents of Ga, N, and GaN using Kozinsky's FLARE++ active learning method.^{24,26} The product of this machine learning approach is a nonparameterized force field that can be mapped into a LAMMPS "pair style." The resulting force field parameters are included in the data availability section of this paper.^{26,27} We trained individual atomic models for GaN, Ga, and N with a B2 descriptor using the FLARE++ on-the-fly learning method.^{24,25} We used a cutoff distance of 4 Å for the pairwise interaction and a normalized dot product kernel raised to the power of two. We then combined the individual atomic models for the three species (GaN, Ga, and N) into a single force field by joining their training sets.²⁶ The final values of the optimized hyperparameters were $\sigma_{\text{var.}} = 4.61$ eV, $\sigma_{\text{force}} = 0.08$ eV/Å, $\sigma_{\text{stress}} = 0.24$ GPa, and $\sigma_{\text{energy}} = 0.37$ eV.

We used an implementation of the FLARE force field within the well-known LAMMPS code for molecular dynamic (MD) simulations.²⁸ We used a time step of 1 fs and an initial substrate size between 1400 and 2048 atoms to balance the computational cost and later added atoms in the bulk phase. We used a Nosé–Hoover thermostat within an isothermal ensemble (NVT) at 2000 K, which corresponds to around 65% of the simulation-derived melting temperature calculated by using the same force field.²⁶ This value of T/T_m was chosen because, as a rule of thumb, this is likely to be close to the maximum stable growth rate, v_{max} .²⁹ The initial crystalline configuration was generated using the Atomic Simulation Environment Python tools,³⁰ and visualization of the MD snapshots was produced using OVITO.³¹

3. RESULTS

Our MD simulations using the FLARE-derived force field were intended to answer a few, as yet unexplored, questions. First, how well does our FLARE-derived force field capture the dynamics of solidification (liquid phase epitaxy) in non-diffusion-limited conditions? In our previous paper, we showed that the combined FLARE model was capable of capturing crystallization from the melt.²⁶ However, the focus of that article was not on the characteristics of the homoepitaxial growth process itself. Hence, in order to give the simulation the best chance of capturing this growth process, we provided a melt phase adjacent to the exposed wurtzite crystal termination. This provided a ready supply of both Ga and N atoms in a stoichiometric ratio.

Second, in this process, nitrogen has to diffuse through liquid Ga in order to create the conditions necessary for GaN solidification; will this be at all limiting? As we show below, we observed a tendency for the two components, Ga and N, to phase-separate during the simulation. This led us to investigate the tendency of N atoms to diffuse through Ga to get to the nascent crystallization front. We would like to uncover the nature of this diffusional process and its role in diffusion-limited crystallization.

Third, how does the growth rate depend on the crystallographic orientation of the GaN surface? Knowledge of anisotropic growth rates has proven invaluable for optimizing vapor-based deposition processes,³² for example, to develop lateral overgrowth methods that can drastically reduce dislocation densities.³³ However, experimental methods, like selective-area growth, cannot be used to measure anisotropic growth rates from the melt, and MD simulations may provide a useful alternate route.

3.1. Crystal Growth from a Stoichiometric, Equimolar, and Well-Mixed Melt. Since we should not expect spontaneous nucleation of GaN crystal from the melt in an

MD simulation,^{34–36} and homogeneous nucleation is not the process of interest here, we seeded the simulation setup with a pre-existing thin film of crystalline GaN. To do so, we set up a "sandwich"-like arrangement for the MD simulations of the crystallization process, as shown in Figure 1a. We created a four-layer thin film with a *c*-plane termination of a GaN crystalline substrate that contained 2048 atoms. We attached a section of vacuum above and below the substrate, extending 30 Å on both sides.

Initially, we used a stoichiometric (50:50) balance of Ga and N atoms in the melt based on an assumption of the ideal availability of both components for crystal growth. We filled the vacuum sections with 2000 gallium atoms and 2000 nitrogen atoms placed randomly in the box, with at least 2 Å of separation between atoms to avoid unwelcome atomic overlaps. We fixed in place the central two layers of the substrate and ran the simulation in an isothermal NVT ensemble at 2000 K for 5 ns. We then used a dislocation analysis^{31,37,38} to characterize the crystallization of any wurtzite structure that formed at the crystal/melt interface. The growth velocity from substrate is defined in eq 1, and the marginal growth velocity is defined in eq 2

$$V_s = (L_i - L_0)/t_i \quad (1)$$

$$V_m = (L_i - L_{i-1})/(t_i - t_{i-1}) \quad (2)$$

where L is the thickness of the crystallized structure. Subscript 0, i , and $i - 1$ indicate the beginning of the simulation, the current time step, and the previous time step, respectively, where each time step is 100 ps.

We plot the growth velocity of crystalline GaN as a function of time in Figure 1c. The initial growth velocity was quite high at 3.37 m/s, but its value quickly decayed after the first 1 ns of the simulation. The average growth velocity over the last 1 ns of a 5 ns MD simulation was only 0.11 m/s, a factor of 30 slower than the original value in the first ns. The growth velocity continued to decrease slowly over time. In order to estimate the growth velocity over a time scale unreachable by MD simulation, we fitted the observed growth velocity to an exponential decay, defined in eq 3.

$$V_t = V_0 \exp(-\lambda t) + V_{\text{inf}} \quad (3)$$

where V_0 is the initial growth velocity, λ is the decay rate with unit ns^{-1} , and V_{inf} is the projected growth rate at times much greater than our simulation time. The fitted value of V_0 was found to be 2.44 m/s, corresponding to the initial rapid growth. The value of V_{inf} was 0.145 m/s, similar to the average growth velocity calculated above, which suggests that we are approaching near-equilibrium crystallization for the simulation. Note that these simulations refer to the case in which there is ample supply of N atoms at the growing interface.

Computationally, this is an unusually slow rate for epitaxial growth in nonreactive systems, e.g., the growth of semiconductors, such as Si, Ge, and their alloy SiGe, or metals,^{29,39,40} which under ideal undercooling conditions can reach the maximum velocity at which it is possible to "pull" the crystal at growth rates up to 15–20 m/s. In comparison, the experimental growth velocity for GaN is exceptionally slow, 0.03–0.7 $\mu\text{m}/\text{min}$ (equivalent to 5×10^{-10} – 1.17×10^{-8} m/s).¹⁵ The experimental growth rate is clearly many orders of magnitude smaller than predicted using the FLARE-predicted force field, given its ideal setup where the supply of N atoms is near-stoichiometric. This difference in growth rates between

our simulation and experiments is likely to be due to the fact that we are observing crystal growth over a time scale 9 orders of magnitude smaller than the experiments.

Returning to the FLARE results, the sharp decrease in growth velocity over the course of the 5 ns simulation time was unexpected; typically, simulations of homoepitaxy proceed at a fairly constant rate. This observation invites the following question: what atomic-scale mechanisms are at play during crystallization? Visualizing the crystallization process suggested a possible explanation. It became clear that some degree of separation had occurred between the gallium and nitrogen species, as shown in Figure 1d. The nitrogen atoms, in particular, tended to form “bubbles” (or clusters) within the melt. This isolated them from their proximity to Ga atoms and hence restricted Ga–N reactions to form the GaN product. Before this segregation had time to occur, namely, at the start of the simulation, when the availability of N atoms at the GaN growth front was high, the growth rate was also high. As segregation of N atoms occurred and their availability at the growth front was greatly diminished, the growth rate plummeted.

We calculated the Ga–Ga radial distribution function (RDF), $g(r)$, in the melt region at three time points in the simulation: after 0, 0.1, and 1 ns, as shown in Figure 1e. Before the simulation started (0 ns), the first peak in $g(r)$ for Ga–Ga in the melt appeared near 3.0 Å, and a second peak appeared near 5.3 Å. Compared to $g(r)$ in liquid gallium, the first peak in a Ga–N mixture was wider and smaller in magnitude. Also, $g(r)$ shifted about 0.4 Å to smaller distances. The difference arose from having both gallium and nitrogen in the melt region before the crystal growth, which reduced the distinctive Ga–Ga first peak. At 0.1 ns, the magnitude of the first peak increased from that at 0 ns, with a similar position for the second peak near 5.3 Å. At a time point of 1 ns, the first peak became more distinct, which corresponds fairly closely to the first peak of the RDF profile in bulk liquid gallium.²⁶ We also calculated the cumulative coordination number of Ga–Ga and N–N interactions in the melt region at these three time points, as shown in the Supporting Information. The coordination number of Ga–Ga in the melt region started at a value close to 4 and increased as phase separation happened. From 0 to 1 ns, the coordination number shifted closer to a value of 12 that corresponds to pure gallium liquid as an ideal hard-sphere liquid. At 1 ns, the coordination number of Ga–Ga in the melt region was 9.5. Meanwhile, the coordination number of nitrogen was 0.7, with the highest peak of $g(r)$ at 1.2 Å and corresponding to the N–N bond length. These changes in the $g(r)$ profile and coordination number in the melt during the crystallization process further supported the existence of a phase separation between the gallium melt and nitrogen species.

However, the two profiles between Ga–Ga interactions during crystal growth and those of pure liquid gallium are not exactly identical: first, the first peak in $g(r)$ has shifted from that of bulk liquid Ga by about 0.2 Å, and the second peak has also shifted by about the same extent. Both peaks in the interfacial system have shifted to slightly shorter distances between Ga atoms compared to those in the bulk liquid Ga. Whether this is a result of the phase separation or is simply within the uncertainty in the peak location of the analysis to create $g(r)$ is unclear. Peak location in $g(r)$ provides a statistical description of local packing (affecting the peak position) and the particle density of the system. This suggests that the

shortening of the Ga–Ga first neighbor distance could be associated with species segregation. Perhaps most noticeable is the loss of structure at and around the second peak, denoted by the much smaller first minimum at around 4 Å and the less well-defined second maximum. These changes are to be expected given the presence of the N atoms, frequently found at second-nearest neighbor distances and beyond and which will prevent Ga atoms from adopting their ideal positions in a bulk Ga liquid. Lastly, to investigate the possibility that the existence of the *c*-plane GaN was responsible for inducing phase separation, we ran additional NVT simulations at 2200 K for a duration of 5 ns using a simulation box measuring 40 × 40 × 60 Å, with Ga/N atom ratios of 2:1 and 3:1. Atoms were randomly placed in the box without the *c*-plane wurtzite GaN substrate. We observed similar phase separation in both Ga–N ratio configurations, but no crystallization occurred without the presence of *c*-plane wurtzite GaN within the time scale of our MD simulations. We include snapshots of these MD simulations and corresponding Ga–Ga and N–N interactions in the Supporting Information.

We suspected that the initial random placement of gallium and nitrogen atoms just above the GaN surface created an artificially well-mixed melt region that might not exist in the experiments. The ready availability of N atoms for the reaction would, therefore, allow the initial crystallization to be much faster. Once phase separation started to occur, the nitrogen atoms had to diffuse through the gallium melt to reach the interface between the crystal growth front and the melt in order to react with the gallium atoms. The supply of nitrogen at the crystal–melt interface thus became limited due to both the rate of N through Ga and of N-clustering (evidence of eventual phase segregation), and the crystal growth slowed down significantly and decayed to about 0.145 m/s. This diffusion-limited mechanism resembled the experimental conditions, wherein the nitrogen source was supplied above the molten gallium droplet on the substrate.

3.2. Diffusional Properties of N in Ga. Since the simulations above have highlighted the importance of N availability near the region of epitaxial growth of GaN, we investigated the transport of N through a pure Ga melt (no GaN seed present). To do so, we simulated nitrogen atoms in a bulk gallium melt with MD to quantify the diffusion behavior of nitrogen in the absence of the crystal–melt interface. We equilibrated the gallium melt within the NPT ensemble at 1 bar for 1 ns and then placed a single nitrogen atom in the system. We analyzed the nitrogen diffusion in Ga over the course of a 10 ns simulation using an NVT ensemble with a Nosé–Hoover thermostat to avoid any volume changes under the barostat.⁴¹

The diffusivity of a single nitrogen atom in a gallium melt was estimated to be between 8.12×10^{-8} and 1.47×10^{-4} cm²/s, as we varied the temperature over a wide range covering 500–2000 K. For comparison, nitrogen self-diffusivity is 2.61 cm²/s at 500 K and 18.75 cm²/s at 2000 K.²⁶ Thus, the diffusion of nitrogen in a gallium melt is 4–5 orders of magnitude slower than nitrogen self-diffusivity, depending on the temperature. Plotting the diffusion coefficient on a logarithmic scale as a function of the inverse temperature, as shown in Figure 2, reveals the expected Arrhenius relationship. The value of our calculated nitrogen diffusivity in Ga was consistent with previous experimental measurements of similar chemical environments: several researchers have estimated the diffusivity of nitrogen in iron and in a silicon melt, as well as the diffusivity of phosphorus in liquid gallium, showing

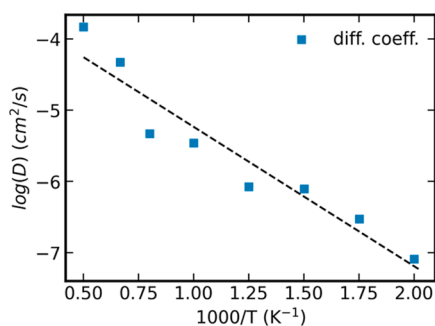


Figure 2. Relation between the diffusion coefficient of nitrogen in a gallium melt and temperature, showing the expected Arrhenius relationship.

diffusivity values between 1×10^{-4} and 1×10^{-6} cm^2/s at temperatures around 1000°C .^{42–44} The diffusivity of nitrogen in Ga was generally low in value across a large temperature range up to the simulated crystallization temperature. As a result of these calculations, it was clear that diffusion is the limiting step for crystallization from the liquid phase during the MD simulation.

3.3. Effect of the Crystal Plane on the Crystal Growth Rate. Turning our attention to more macroscale concerns, it is generally of interest to know the relative growth rates among the different crystal facets. To consider this, we compared the rate of crystallization on three different planes: *a*-, *m*-, and *c*-planes. The polar *c*-plane is the most commonly grown plane

because of its lattice stability and its wide suitability, given that it is lattice-matched to common substrates such as sapphire.^{32,45} However, *c*-plane GaN is also susceptible to spontaneous and strain-induced polarization.^{46,47} Nonpolar planes, such as *a*- and *m*-planes, can mitigate the polarization in GaN heterostructures and increase quantum efficiency in optoelectronics applications.^{48–51} We compared the simulated crystal growth on different growth planes to look for differences between polar and nonpolar liquid-phase growth behaviors, again using our FLARE-generated force field and modeling design.

To investigate this facet dependence, we ran two sets of crystallization simulations. In the first set of simulations, our approach was to constantly replenish the melt region with gallium and nitrogen atoms on both sides of the growth front (illustrated in Figure 1a). Atoms were added every 200 ps with a minimum distance of 2 Å away from nearby neighbors to avoid unwanted atom–atom overlap. The idea behind these simulations was to maintain a well-mixed environment in the melt region and circumvent the diffusion-limiting steps that we had observed previously so that we could achieve a near-saturated growth rate in the most ideal situation.

The second set of simulations took a different approach: here, we created only the melt region on one side of the growing GaN crystalline substrate plane, as illustrated in Figure 1b. The bottom layer of the substrate was fixed during the simulation, and a repulsive wall was placed in the *z*-direction to prevent atoms from crossing the periodic boundaries. We then

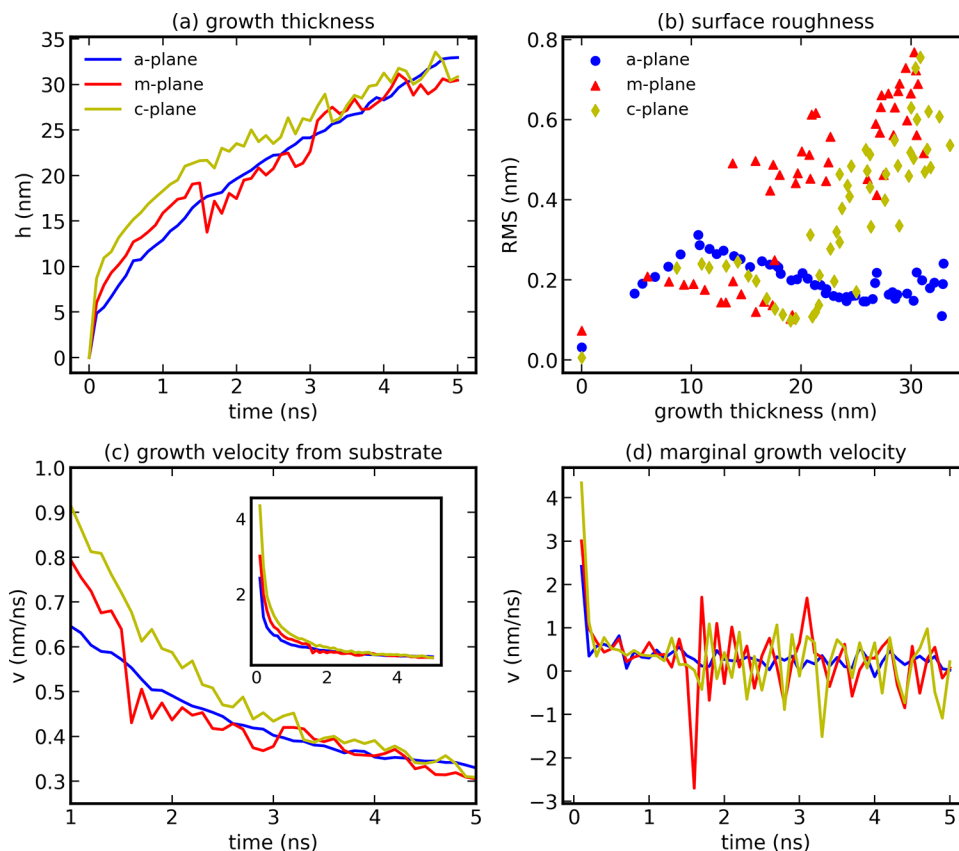


Figure 3. Crystal growth profiles of three different exposed facets (described and color-coded in the insets) in the “sandwich”-like system setup, as illustrated in Figure 1a (shown in the four panels of (a) at the top of the figure). Gallium and nitrogen atoms were replenished every 200 ps in the melt region with a minimum distance of 2 Å from nearby neighbors. The four panels in each case show the growth thickness, surface roughness, growth velocity, and marginal growth velocity, each as a function of simulation time.

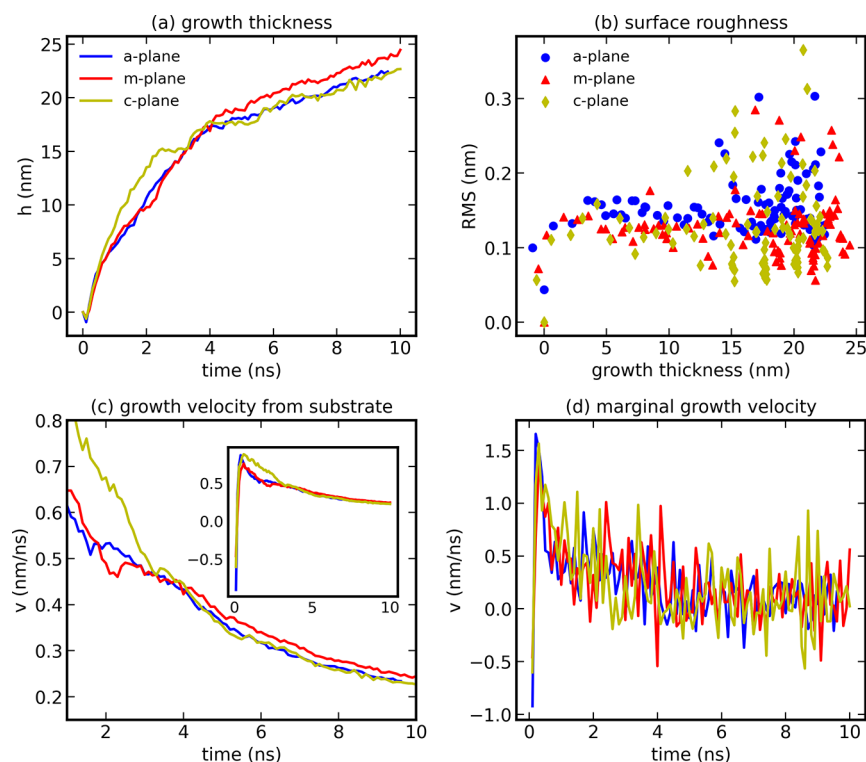


Figure 4. Crystal growth profiles for three different exposed facets (described and color-coded in the insets) in a nonperiodic setup, as illustrated in Figure 1b [four panels of (b) at the top of the figure]. Nitrogen atoms were replenished every 200 ps to maintain a nitrogen overpressure. The four panels in each case show the growth thickness, surface roughness, growth velocity, and marginal growth velocity, each as a function of simulation time.

filled the vacuum with gallium and nitrogen atoms at a 3:1 Ga/N ratio to make sure that gallium is not the limiting species and allow nitrogen diffusion from above. This setup allowed us to incorporate the nitrogen diffusion mechanism into the overall crystallization. We also replenished the supply of nitrogen atoms above liquid gallium, again with a minimum interatomic distance of 2 Å between atoms introduced every 200 ps. This was done to maintain the nitrogen overpressure and guarantee the occurrence of nitrogen diffusion toward the growth front within the time scale of MD simulations. In both sets of simulations, we characterized the crystallized layer's thickness, growth velocity, and surface roughness for each of the three planes we investigated; see Figures 3 and 4. To characterize the surface roughness, we divided the growth front into a 2 by 2 Å grid. Surface roughness was defined as the root mean squared average (RMS) of the growth front height deviation from the arithmetic average during the growth.

As mentioned earlier for the *c*-plane, in the near-saturated growth (method 1), the initial growth rates were 3–4 m/s on all three planes that quickly decreased to 0.6–0.9 m/s after the first 1 ns. After 5 ns, the average growth velocity was around 0.35 m/s, and there was no significant difference between growth on the different planes. Most of the atoms in the melt region have crystallized within the first 5 ns, hence the decision to compare trajectories over this time scale. Judging from the fluctuations in the marginal growth velocity, the *a*-plane had the steadiest growth, followed by that of the *c*-plane. Growth on the *m*-plane was the least steady. Correspondingly, the *m*-plane exhibited the roughest surface, with RMS $\approx$ 0.8 nm even shortly after about a 10 nm thickness. In contrast, the *a*-plane had the smoothest surface growth.

In the diffusion-limited growth simulation runs (method 2), the overall growth profiles were different. Although each plane had a negative growth velocity initially and seemingly a very quick “decomposition” period, this was due to disordered atoms at the interface at the start of the crystallization. Because the crystallization process was in constant dynamic flux, the initial growth velocity showed a decrease and was similar to the oscillating marginal growth velocity in Figure 4. The growth velocity then increased to around 1 m/s and slowly decayed to 0.2–0.3 nm at 10 ns. Although the differences in growth rates as a function of the exposed facet were only small, only around 10%, the *m*-plane had the highest growth velocity compared to those of the *a*- and *c*-planes. Similar to the results from the simulations using the first method, the *m*-plane has relatively less steady growth and hence has created a less smooth surface over the course of the crystallization process as compared to the other two planes.

For these diffusion-limited simulations, we again fitted the growth rate decay to an exponential decay, as defined in eq 3. We included the growth decay parameters in Table 1. In this more experimentally applicable simulation setup, the value for V_{inf} is far smaller, on the order of 10^{-4} m/s. Overall, the *a*- and *c*-planes exhibited more steady growth behavior and produced

Table 1. Exponential Decay Parameters for *a*, *c*, *m* Planes During Diffusion-Limited Growth Simulations

plane	V_0 (m/s)	λ (ns $^{-1}$)	V_{inf} (m/s)
<i>a</i> -plane	0.71	0.13	4.27×10^{-4}
<i>c</i> -plane	0.86	0.17	1.41×10^{-4}
<i>m</i> -plane	0.67	0.11	1.49×10^{-4}

smoother crystal planes than growth on the *m*-plane. The difference in growth rate and surface smoothness among the planes was only small, as generated from our MD simulations, and probably not significant. There are ramifications of our observation that nitrogen diffusion was the limiting step in the liquid phase epitaxy of GaN, regardless of the exposed facet. This creates a high gallium/nitrogen ratio at the crystallizing front. By the time nitrogen atoms can diffuse to the growth front, nitrogen that has already reached the growth front has crystallized. This slow rate of approach (and hence crystallization) results in the formation of a relatively smooth surface at the interface. Similar results have been observed by others, including that the ratio of N to Al affected crystal growth quality.^{52,53}

Lastly, we compared the surface energy of the *a*-, *c*-, and *m*-planes from the MD results. In our previous study, the FLARE-generated force field was able to calculate the elastic constants and lattice constants of GaN as accurately as those from density functional theory (DFT) calculations and experimental measurements.²⁶ Encouraged by this result, we ran a surface energy scan in LAMMPS at 0 K, which, hopefully, should resemble a DFT energy scan. We defined the surface energy, $\sigma_{\text{surface}} = E_{\text{surface}} - N \times E_{\text{Ga-N pair}}$, where *N* is the number of total atoms in the system divided by two. We created at least 20 layers of each plane to approximate a semi-infinite slab. Vacuum occupied the direction perpendicular to the plane to a thickness of 10 Å, and the direction parallel to the plane lay across the periodic boundary. In the *c*-plane, this method estimated the average surface energy between the N-terminated and Ga-terminated planes because of the polarity of the *c*-plane. The resulting surface energy values for the three planes were all very close: 0.0631 eV/Å² for the *a*-plane, 0.0581 eV/Å² for the *c*-plane, and 0.0731 eV/Å² for the *m*-plane. The *m*-plane had the (marginally) highest surface energy, which corresponded to the least stable growth in our liquid-phase epitaxy simulation. On the other hand, the *a*-plane surface energy was 9% higher than the *c*-plane, which resulted in the most stable growth in our previous simulations. However, these small differences in surface energies were overshadowed by the impact of the diffusion-limited process of nitrogen migration toward the interface.

4. DISCUSSION

In this work, we took advantage of the availability of a machine-learned FLARE force field to simulate and analyze GaN crystal growth.²⁶ Initially configured in a sandwich-like system setup in which there was a stoichiometric ratio of Ga/N, we observed rapid crystal growth at the solid/liquid interface in the first 1 ns. Thereafter, the growth velocity quickly decayed to a value around 0.11 m/s and still fell slowly as the simulation proceeded. We extrapolated the growth rate, assuming exponential decay, to a value of V_{inf} of around 0.145 m/s, suggesting that we were close to a steady-state value after about 5 ns. During this process, we observed a phase separation of gallium and nitrogen species within the melt region after the initial 1 ns. We hypothesized that the initial rapid growth was caused by the arbitrarily well-mixed melt region from the simulation setup and that the subsequent phase separation slowed the crystallization.

To prove this hypothesis, we showed that the RDF of gallium in the melt region after 1 ns resembled that calculated for pure liquid gallium. Moreover, we quantify the nitrogen diffusion in liquid gallium to be at least 4 orders of magnitude

lower than the nitrogen self-diffusion at 1000 K. The mass transfer of nitrogen from the surface of liquid gallium to the substrate/liquid interface was the limiting step in the crystallization of the GaN liquid phase epitaxy process.

We then reconfigured the simulation setup to more closely resemble that in the experiments and also ran simulations exposing three different planes, *a*-, *c*-, and *m*. Placing nitrogen atoms above the liquid gallium region, through which they had to diffuse, allowed us to incorporate a more realistic diffusion-limited mechanism into the MD simulations. We compared the growth velocity and surface roughness among the three prevalent planes. In this scenario, as expected, we did not observe very rapid growth at the beginning compared to the well-mixed system setup. Again assuming exponential decay, the growth velocity decayed to an extrapolated value around 1.5×10^{-4} m/s, which corresponded much more closely to the slow growth observed in the additive manufacturing experiments (on the order of 10^{-8} m/s). Our FLARE-generated growth rates were far smaller than are typically found in MD simulations of liquid phase epitaxy by orders of magnitude. This is encouraging, in terms of the accuracy and usefulness of the FLARE predictions for liquid phase epitaxy. The *a*- and *c*-planes showed steadier growth than that on the *m*-plane. The three planes gave virtually indistinguishable growth rates, so we could not identify the plane with the lowest growth velocity. This may be due to nitrogen diffusion being the rate-limiting step.

Overall, our work shows the value of a machine-learned force field to offer a compelling compromise between the accuracy of ab initio MD and the system size and time scales accessible using a traditional classical MD approach. This opens up the opportunity of conducting more sophisticated MD simulations, such as investigating the temperature dependence of crystal growth and estimating the interfacial free energy between the crystal and bulk liquid phases during crystallization.

■ ASSOCIATED CONTENT

Data Availability Statement

Materials and Correspondence: Correspondence should be addressed to Paulette Clancy at pclancy3@jhu.edu and materials request should be addressed to Xiangyu Chen at xchen150@jhu.edu.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.cgd.3c01504>.

RDF and cumulative coordination number of Ga–Ga and N–N interactions during the phase separation and simulation snapshot of only the gallium and nitrogen mixture (PDF)

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Author Contributions

X.C., N.Q.L., and P.C. were engaged in the design of the project. X.C. conducted the computations and analysis. X.C. wrote the manuscript in conjunction with N.Q.L. and P.C. P.C. supervised the project and provided critical feedback.

Notes

The authors declare no competing financial interest.

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